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BREATHETWIN

TEAM MOMENTUM

EARLY WARNING. EASIER BREATHING.

BreatheTwin

A COPD digital twin for India

Warns 1 to 3 days before a flare-up, explains why, and shows what to do

Team MOMENTUM · SDA Bocconi Asia Center, Mumbai

Digital Twin Challenge 2026 · Happiest Health

AT

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Flare-ups warn days ahead, yet India catches them too late

India's COPD burden is large

- **55 million** people in India lived with COPD in 2016 *(India State-Level Disease Burden Initiative, Lancet Global Health 2018)*
- Chronic respiratory disease caused **10.9%** of all deaths in India in 2016 *(same study)*

Flare-ups are the dangerous moments

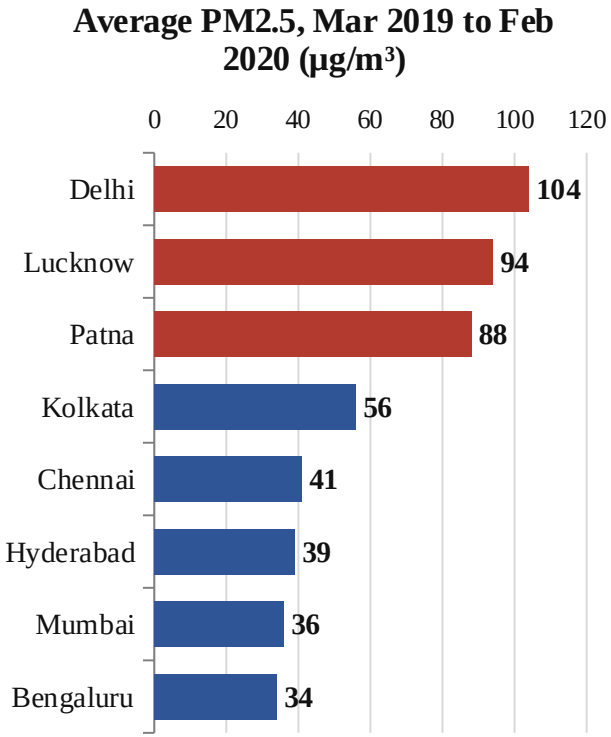
- Patients have about **1 to 2 flare-ups a year**, more as COPD worsens *(ECLIPSE, NEJM 2010)*
- Warning signs start **5 to 7 days** earlier: breathing rises by **4.4 breaths/min** the day before *(Yañez 2012; Hurst 2010)*
- Today they are noticed only when the patient is already struggling

The Care Gap

Care today	What patients need
Check-ups every few months	A daily watch against their own normal
Acts after symptoms appear	A warning 1 to 3 days ahead
Same alarm limits for everyone	Limits personal to each patient
Ignores city air and weather	Record, body and air in one view

Air makes it worse *(real CPCB data, 2019-20)*

- Delhi PM2.5 averaged **104 µg/m³**, about **7×** the WHO daily safe level of 15
- Winter air was **3× dirtier** than monsoon air (104 vs 32 µg/m³)



The opportunity:

Use the days of warning the body already gives: a personal twin that spots a flare-up early, explains why, and tells families what to do.

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BreatheTwin: a twin that warns 1 to 3 days before a flare-up

Core Proposition

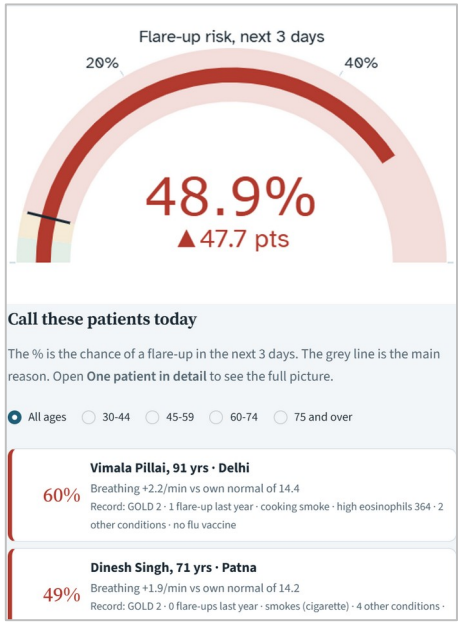
A personal twin for every COPD patient that:

- Learns each person's **own normal**
- Predicts a **flare-up 1 to 3 days ahead**
- **Explains the reasons** in plain words
- Shows what would **lower the risk**
- Gives families **smog warnings** and a **visit summary** for the doctor

3 Data Streams

Stream	What it adds
Health record (EHR)	Severity, past flare-ups, smoking, blood tests, other illnesses: the *starting risk*
Wearable + smart inhaler	Daily heart rate, breathing, oxygen, steps, sleep, inhaler use: the *early warning*
City air + weather	PM2.5, cold snaps, humidity: the *seasonal trigger* and the advice

Tech & Experience



Doctor's view: the risk, who to call today, and why

Key Differentiator

Usual care	Wearables	BreatheTwin
Reactive	Raw numbers	**Predictive**
Textbook limits	Textbook limits	**Own normal**
No air data	No record	**Record + body + air**
Doctor only	User only	**Doctor + family**

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Works with the wearables and IoT devices patients can already buy

Which sensor tracks which signal

Signal	Sensor	Devices	Twin uses it as
Resting heart rate	Optical PPG (green light)	Any smartwatch, band, oximeter	Change vs own normal
Breathing rate	PPG + motion, during sleep	Fitbit, Apple, Galaxy, Garmin watch	Change vs own normal
Blood oxygen (SpO2)	Red + infrared light	Fingertip / ring oximeter, watch	Change; urgent below 90%
Steps / activity	Accelerometer	Watch, band or phone	Change vs own normal
Sleep hours	Accelerometer + heart rate	Watch or band	Change vs own normal
Rescue puffs, doses	Clip-on actuation sensor	ADHERO (Lupin), Propeller, Hailie	Puffs vs normal; adherence
Blood pressure	Cuff pressure (oscillometric)	Omron Bluetooth monitor	Shown to the doctor
PM2.5, temperature	CPCB stations, weather API	City networks (no device)	Dose-response, cold snaps

Three ways to equip a patient

Minimum kit: Any watch that syncs to Health Connect or Apple Health + fingertip oximeter. Gives heart rate, breathing, SpO2, steps, sleep.

Full kit: + smart inhaler clip (ADHERO, launched in India by Lupin) + Omron BP cuff. Adds automatic inhaler tracking.

No-watch kit: Phone + oximeter; the family types SpO2 and a 30-second breathing count. Near-zero cost.

No needle, no blood test, nothing under the skin:

a watch and a clip-on oximeter carry the daily signal; data flows phone → Health Connect / HealthKit → BreatheTwin.

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How the twin works: three data streams, one personal model

Data In

Health record (EHR)
400 synthetic patients: Synthea + India overlay

Wearable + smart inhaler
heart rate, breathing, SpO2, steps, sleep, puffs, doses

City air + weather
real CPCB PM2.5 and Open-Meteo, 8 cities

The Twin

1. Personal baseline
Every reading compared with the patient's **own last 4 weeks**, not a textbook range

2. Hybrid risk engine
LightGBM learns body signals, inhaler use, weather and record × **published PM2.5 effect** (DeVries 2017)

What Comes Out

- **Risk of a flare-up in the next 3 days**
- **Reasons, ranked by how much they push risk**
- **Safety checks against own normal**
- **What-if answers for doctor and family**
- **Smog warnings + one-page visit summary**

Twin journey

Health record → **Daily readings** → **Own normal** → **Risk score** → **Reasons** → **Action + advice**

Why hybrid: the body-signal patterns are complex and best learned from data; the air-pollution effect is small and well measured in large studies, so we take it from the literature.

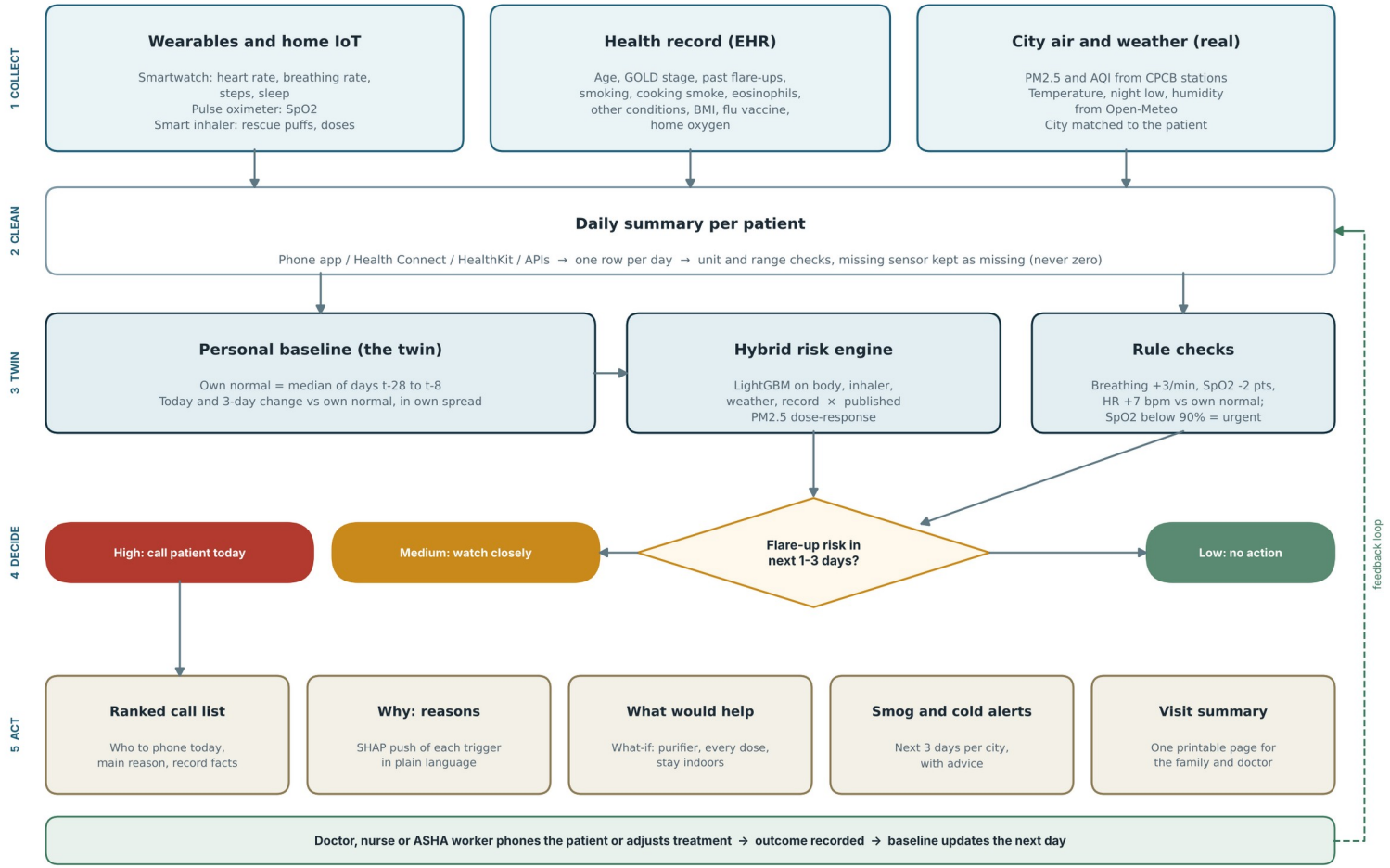
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Flowchart: what BreatheTwin does for every patient, every morning



1 Collect. Watch, oximeter, inhaler clip; health record; real city air and weather.

2 Clean. One row per patient per day; range checks; missing stays missing.

3 Twin. Today vs own normal (days t-28 to t-8), hybrid risk engine, rule checks.

4 Decide. High: call today. Medium: watch. Low: no action.

5 Act. Call list, reasons, what-if, smog alerts, visit summary; outcome feeds back.

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Open, anonymised or synthetic data, and clear about which

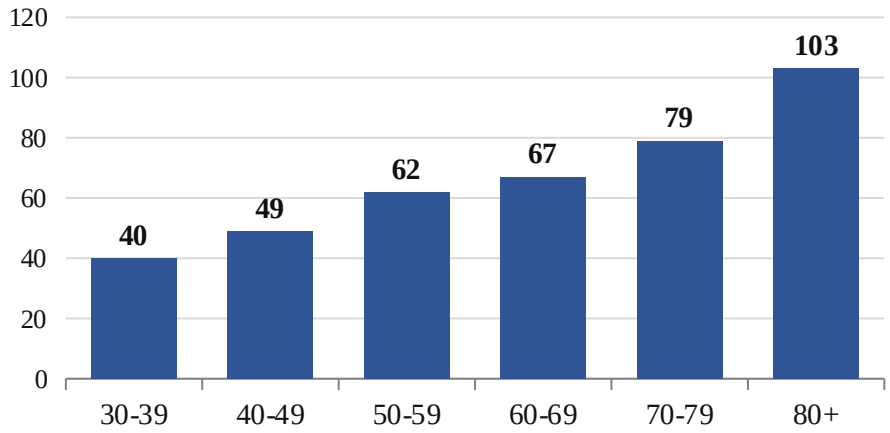
Data Sources

Stream	Source	Real or synthetic	Licence
Health records	Synthea COPD cohort + our India overlay and younger patients	Synthetic	Apache 2.0
Wearables and flare-ups	Simulated; every effect size from a published study	Synthetic	Ours (MIT)
Air quality	CPCB daily PM2.5, 8 cities (Kaggle 'Air Quality Data in India')	**Real**	CC0
Weather	Open-Meteo archive: temperature, humidity	**Real**	CC BY 4.0
Real-data check	MIMIC-IV Clinical Database Demo v2.2 (14 COPD patients)	**Real**, aggregate only	ODbL 1.0

Real-data check: sex mix and heart disease match real patients; heart failure and diabetes are lower in synthetic records.

Our Patient Cohort (400)

Patients by age group: every adult generation



- Severity GOLD 1 / 2 / 3 / 4: **17% / 42% / 29% / 12%**
- Current smokers **18%** · cooking smoke at home **31%** · 8 cities

Honest by design:

Air and weather are real; patients, watch readings and flare-ups are simulated, so the results prove the idea works, not clinical accuracy.

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How much of our data is real, and what each part is used for

Every data stream, real or synthetic

Data	Real?	Size	Used for
City air quality (PM2.5, AQI)	Real	2,928 city-days, 8 cities (CPCB)	Training + dashboard
Weather (temperature, humidity)	Real	2,928 city-days (Open-Meteo)	Training + dashboard
Health records	Synthetic	400 patients (Synthea + India overlay)	Training + dashboard
Watch, oximeter, inhaler readings	Synthetic	146,400 patient-days, from studies	Training + dashboard
Flare-ups	Synthetic	473 simulated events	Training labels
Real hospital patients (stress test)	Real	100 patients, 47,214 vital readings	Testing only
Real COPD records (check)	Real	14 COPD patients (MIMIC-IV demo)	Testing only

100%

of air and weather data

is real (CPCB, Open-Meteo)

~1%

of all training values

are real; patient data is simulated

100

real patients, 55 women, 45 men

aged 18 to 90, used to stress-test

Why simulate patients at all?

No open dataset links COPD wearable data to labelled flare-ups, so we simulate from published studies and test on real people.

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AI model: learn the body, borrow published science for the air

What the Model Looks At

Data input	Signal the twin uses
Breathing, oxygen, heart rate	Change vs **own normal** (today, 3-day average, how unusual)
Steps and sleep	Drop vs own normal
Inhalers	Extra rescue puffs; % of daily doses taken this week
Weather	3-week temperature drop, night low, humid heat
Health record	Age, GOLD stage, past flare-ups, smoking, cooking smoke, eosinophils, other illnesses, BMI, flu vaccine, home oxygen
Air (PM2.5)	Published dose-response, **not learned**: +2.5% risk per 10 µg/m³

Tech stack: Python · pandas · LightGBM · scikit-learn · Streamlit · Plotly · Synthea (Java)

Open source: MIT licence; whole pipeline rebuilds with one command

Model Details

Algorithm	LightGBM (gradient-boosted decision trees)
Predicts	Will a flare-up start in the next 1 to 3 days?
Tested on	120 patients never seen in training (280 trained)
Safety rules	Warning signs can never lower risk; more inhaler doses can never raise it
Explanations	Per-factor contributions (SHAP values): "what is pushing the risk"

Every number has a source

- Breathing rise before flare-ups: *Yañez 2012*
- Flare-up rates by severity: *ECLIPSE, Hurst 2010*
- Eosinophils ≥340/µL: **1.76×** flare-ups *(Vedel-Krogh 2016)*
- Still smoking: **1.28×** *(Au 2009)* · PM2.5: *DeVries 2017*
- Anything unsourced is labelled **ASSUMPTION** in the code

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Results: 2 in 3 flare-ups flagged 1 to 3 days ahead

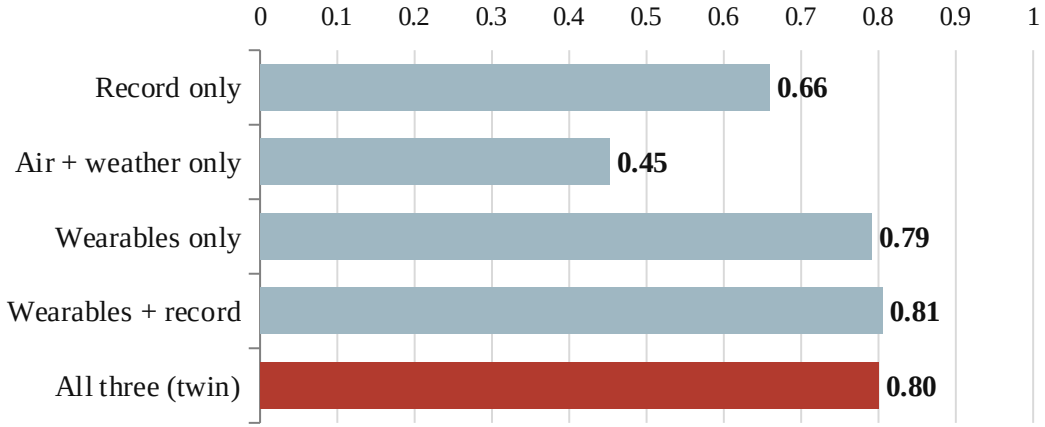
0.80
Accuracy score (AUROC)
published tools: 0.68 to 0.85

66%
Flare-ups warned
1 to 3 days ahead

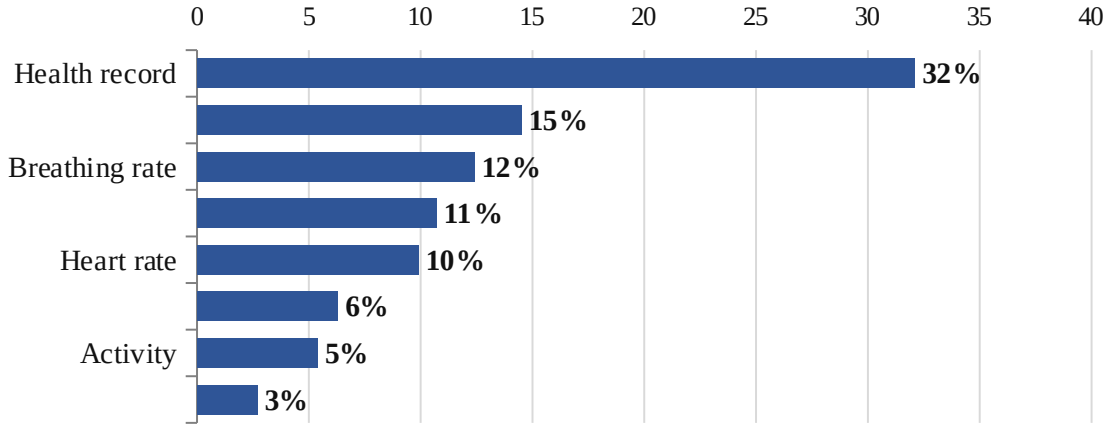
0.8
False alerts
per patient per month

120
Unseen test patients
never used in training

Prediction quality by data stream (AUROC)



Share of the twin's reasoning by factor (%)



What this means:

Body signals carry the early warning; the health record sets each person's starting risk; air and weather explain the seasons.

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Stress test on 100 real patients: robust, fair, right direction

75/100
real patients assessed
25 had under 48 h of vitals

0
crashes or invalid risks
incl. missing oxygen sensor

0.71
risk vs real worsening
rank correlation

Pass
worse readings raise risk
8-case edge suite

By sex and age group

Group	Patients	Median risk	Alerted
Female	44	0.9%	30%
Male	31	0.6%	19%
18-39	2	20.2%	50%
40-49	6	0.6%	17%
50-59	7	0.6%	29%
60-69	9	0.6%	0%
70-79	18	0.6%	22%
80+	32	0.8%	31%

What we learned

Robust: every real patient with 48 h of data was scored; no crash, no impossible risk.

Right direction: breathing up to 35/min gives High risk; all three signs worse gives 96%.

Age and sex: age 18 vs 90 and switching sex barely change risk; sex is not a model input.

To fix: an SpO2 drop alone barely moves the learned risk (the rule layer still flags it urgent).

Limits: no link with later in-hospital death (AUROC 0.49); women alerted more at equal signs, but groups are 5 to 19 people.

Real people, never used for training:

MIMIC-III demo v1.4 (PhysioNet, ODbL). ICU patients are not a COPD home cohort: a stress test, not validation.

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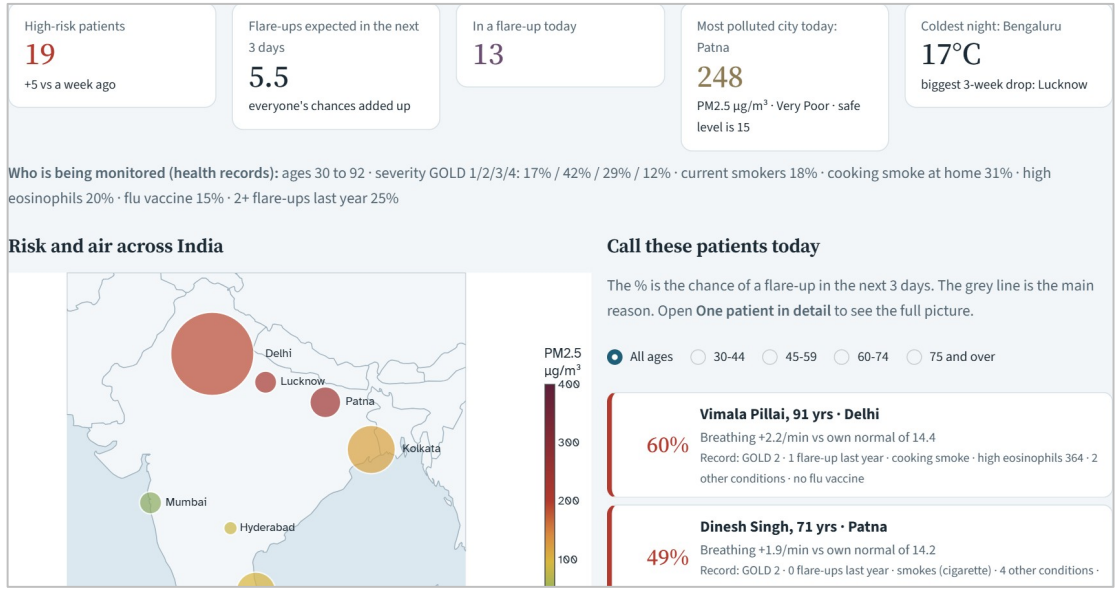
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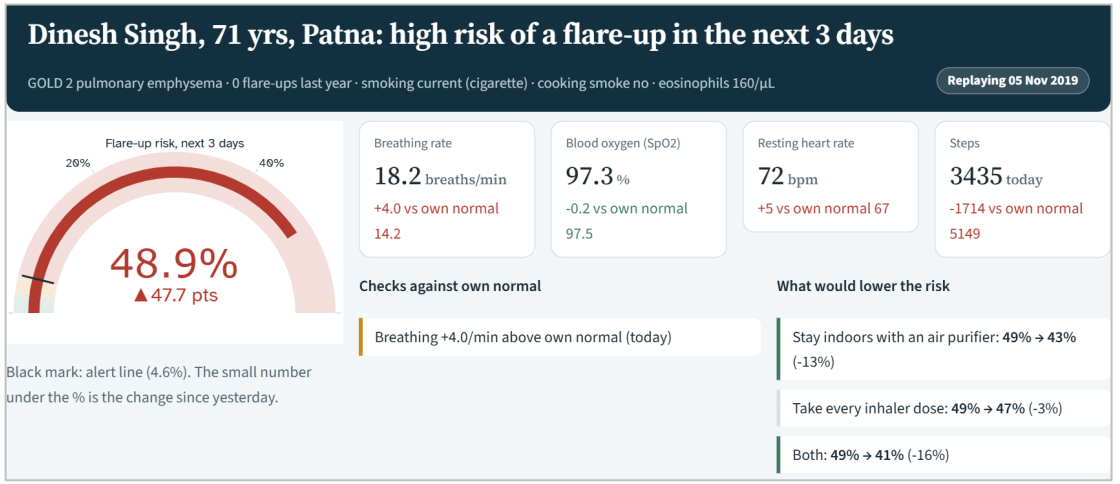
The doctor's dashboard: who to call today, and why

Overview: all 400 patients



- **Call these patients today**: ranked by risk, with the main reason and key record facts
- India map of risk and air; filter by age group; plain-language guide on every page

One patient in detail



- Risk dial with the alert line and the change since yesterday
- Today's readings vs **own normal**: red means worse than usual
- **What would lower the risk**: e.g. air purifier 49% → 43%

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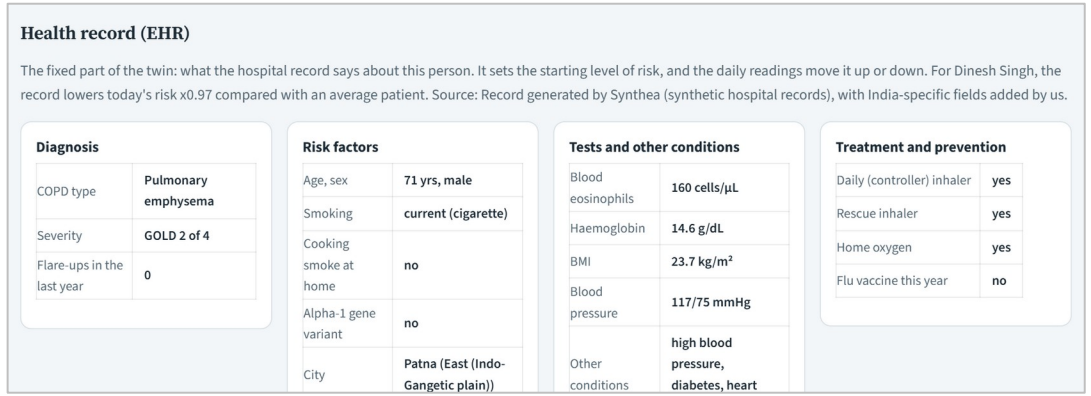
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Health record in every view, and a lab to test advice before giving it

Health Record (EHR)



Health record (EHR)

The fixed part of the twin: what the hospital record says about this person. It sets the starting level of risk, and the daily readings move it up or down. For Dinesh Singh, the record lowers today's risk x0.97 compared with an average patient. Source: Record generated by Synthea (synthetic hospital records), with India-specific fields added by us.

Diagnosis

COPD type	Pulmonary emphysema
Severity	GOLD 2 of 4
Flare-ups in the last year	0

Risk factors

Age, sex	71 yrs, male
Smoking	current (cigarette)
Cooking smoke at home	no
Alpha-1 gene variant	no
City	Patna (East (Indo-Gangetic plain))

Tests and other conditions

Blood eosinophils	160 cells/µL
Haemoglobin	14.6 g/dL
BMI	23.7 kg/m²
Blood pressure	117/75 mmHg
Other conditions	high blood pressure, diabetes, heart

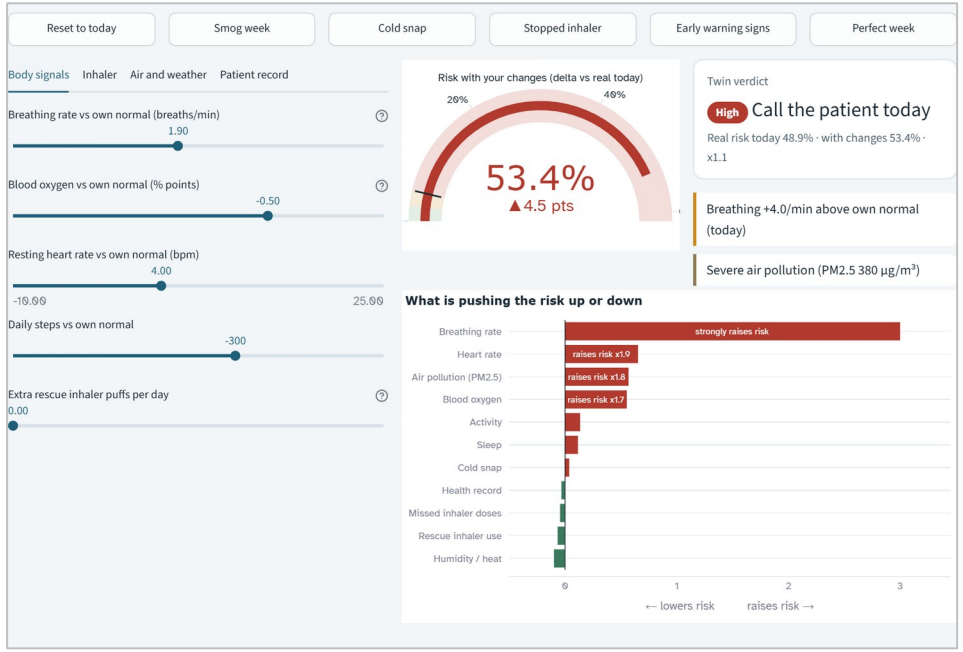
Treatment and prevention

Daily (controller) inhaler	yes
Rescue inhaler	yes
Home oxygen	yes
Flu vaccine this year	no

The fixed part of the twin

- Diagnosis, risk factors, blood tests, other illnesses, medicines
- Shows how much the record **raises or lowers** today's risk
- Record facts also appear on the call list, what-if lab and new-person check

What-if Lab



- One click scenarios: **smog week, cold snap, stopped inhaler, early warning signs, perfect week**
- Sliders for body signals, inhaler, air, weather and the record; risk re-scores instantly

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Check a new person in minutes, and see every value the twin used

Result for a new person with warning signs



How it works

1. Health record: age, severity, smoking, blood tests
2. Readings **usual** (a good day) and **now** (last 3 days)
3. Optional: today's PM2.5 from a CPCB or AQI app
4. Risk, reasons, safety checks and a table of every value used

We tested it

Test case	Result
Readings now = usual	**Low**, about 0.4%
Faster breathing, lower oxygen, more puffs	**High**, about 96%
Eosinophils 220 → 520 cells/µL	Risk 0.43% → **0.80%**

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From prediction to action: smog warnings and a visit summary

Smog is Coming: next 3 days

Smog and cold warnings: next 3 days

⚠ **Lucknow: air expected to be very poor on Wednesday, Thursday, Friday (PM2.5 up to 221 µg/m³).**

- Stay indoors as much as possible and keep windows closed
- Wear an N95 mask if you must go out
- Keep the rescue inhaler with you and take the daily inhaler without missing doses
- Avoid other smoke at home: incense, mosquito coils, wood or kerosene cooking

At this level of PM2.5, a COPD patient's flare-up risk over the following week is about **57% higher** than on a clean-air day (published effect).

⚠ **Patna: air expected to be very poor on Wednesday, Thursday, Friday (PM2.5 up to 202 µg/m³).**

- Stay indoors as much as possible and keep windows closed
- Wear an N95 mask if you must go out
- Keep the rescue inhaler with you and take the daily inhaler without missing doses
- Avoid other smoke at home: incense, mosquito coils, wood or kerosene cooking

At this level of PM2.5, a COPD patient's flare-up risk over the following week is about **49% higher** than on a clean-air day (published effect).

⚠ **Delhi: air expected to be very poor on Wednesday, Thursday, Friday (PM2.5 up to 182 µg/m³).**

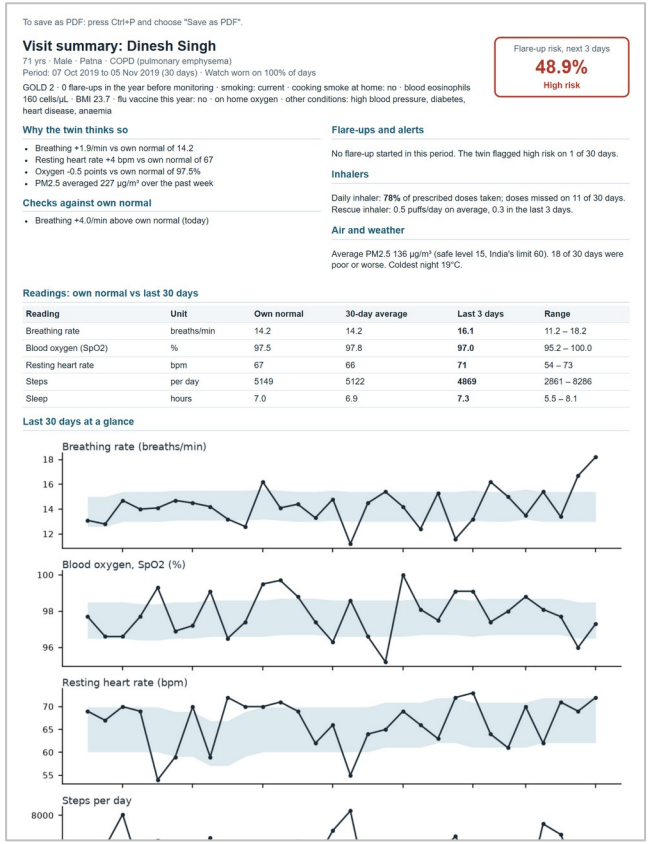
- Stay indoors as much as possible and keep windows closed
- Wear an N95 mask if you must go out

⚠ **Chennai: air expected to be very poor on Wednesday, Thursday, Friday (PM2.5 up to 143 µg/m³).**

- Stay indoors as much as possible and keep windows closed
- Wear an N95 mask if you must go out

- Uses India's official **CPCB PM2.5 bands**; advice gets stronger as air gets worse
- Also warns about **cold nights**; shows the extra flare-up risk from the published effect
- *Demo replays real 2019-20 data; a live version would use an air-quality forecast*

One-page Visit Summary



To save as PDF: press Ctrl+P and choose "Save as PDF".

Visit summary: Dinesh Singh

71 yrs · Male · Patna · COPD (pulmonary emphysema)
Period: 07 Oct 2019 to 05 Nov 2019 (30 days) · Watch worn on 100% of days

Flare-up risk, next 3 days: **48.9%** High risk

GOLD 2 · 0 flare-ups in the year before monitoring · smoking: current · cooking smoke at home: no · blood eosinophils 160 cells/µL · BMI 23.7 · flu vaccine this year: no · on home oxygen · other conditions: high blood pressure, diabetes, heart disease, anaemia

Why the twin thinks so

- Breathing +1.9/min vs own normal of 14.2
- Resting heart rate +4 bpm vs own normal of 67
- Oxygen -0.5 points vs own normal of 97.5%
- PM2.5 averaged 227 µg/m³ over the past week

Checks against own normal

- Breathing +4.0/min above own normal (today)

Flare-ups and alerts

No flare-up started in this period. The twin flagged high risk on 1 of 30 days.

Inhalers

Daily inhaler: **78%** of prescribed doses taken; doses missed on 11 of 30 days. Rescue inhaler: 0.5 puffs/day on average, 0.3 in the last 3 days.

Air and weather

Average PM2.5 136 µg/m³ (safe level 15, India's limit 60). 18 of 30 days were poor or worse. Coldest night 19°C.

Readings: own normal vs last 30 days

Reading	Unit	Own normal	30-day average	Last 3 days	Range
Breathing rate	breaths/min	14.2	14.2	**16.1**	11.2 – 18.2
Blood oxygen (SpO2)	%	97.5	97.8	**97.0**	95.2 – 100.0
Resting heart rate	bpm	67	66	**71**	54 – 73
Steps	per day	5149	5122	**4869**	2861 – 8286
Sleep	hours	7.0	6.9	**7.3**	5.5 – 8.1

Last 30 days at a glance

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Who it helps: doctors, health workers and families

Value by Stakeholder

Who	Today	With BreatheTwin
Doctor / clinic	Sees the patient only when they come in sick	A ranked daily list of who to call, with reasons
Community health worker	Visits by rota, not by need	Visits the few patients at highest risk first
Patient and family	Notice a flare-up when breathing is already bad	Early warning, smog advice, a summary to take to the doctor
Health system	Emergency visits in smog season	Aim: fewer emergency visits through early action

Why it fits India

- Built for **Indian triggers**: cooking smoke, bidis, winter smog
- Covers **ages 30 to 92**, since COPD starts earlier in India
- **Plain language** on every page, with a glossary
- **Open source** and free to adapt
- Planned: low-cost pulse oximeter + phone, Hindi

The care loop

Watch daily → **Warn early** → **Act at home** → **Visit with summary** → **Learn and refine**

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Honest limits, and the road from prototype to real patients

Limitations and How We Handle Them

Limitation	How we handle it
Patients, readings and flare-ups are simulated	Every effect sourced or labelled ASSUMPTION; real air and weather; stress-tested on 100 real patients
No India-specific PM2.5 effect for COPD	Use the global published figure; flag it clearly
Synthetic records differ from real ones	Checked against real MIMIC-IV patients; run on 100 real MIMIC-III patients; gaps reported
Demo replays past air data	Same rules plug into a live forecast
Needs a smartwatch or oximeter	Minimum kit is a watch + oximeter; no-watch kit uses typed readings

Roadmap

Phase 1: Validate

Partner hospital · test on real Indian COPD records · tune alert line

Phase 2: Patient app

"My lungs today" · daily symptom check-in · SMS / WhatsApp alerts · Hindi

Phase 3: Scale

Live air-quality forecasts · link to ABDM health records · health-worker rollout

Next step:

Validate on real Indian COPD patients with a hospital partner before any clinical use.

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Thank You!

Anushka Tomar (Team leader) · Yash Saxena

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